Detection Without Identification: A Validity Assessment of the ShotSpotter Difference-in-Differences Design in Chicago, 2009–2023

Austin Belman and Kameron Nelson

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Abstract

A two-way fixed-effects difference-in-differences regression of Chicago gun homicides on ShotSpotter installation returns a positive, statistically significant estimate of $+2.63$ homicides per community-area-year ($p < 0.001$). We show that the estimate is an artifact of two choices rather than a treatment effect. Modeling integer homicide counts with ordinary least squares inflates the coefficient; estimated by Poisson or Negative Binomial regression with the same fixed effects, the effect falls to an incidence-rate ratio of 1.07 (95% CI $[0.87, 1.33]$, $p = 0.51$), a statistically null result. The design also fails the conditions difference-in-differences requires. A joint Wald test rejects parallel pre-trends ($F = 19.9$, $p = 0.006$). Treatment dates assigned to pre-rollout years (1999, 2003) produce coefficients of comparable magnitude and significance to the post-2017 estimate. Synthetic control is infeasible because the 26 never-treated community areas are lower-violence than any treated neighborhood, so no donor pool reproduces a treated unit. The Callaway and Sant'Anna (2021) estimator returns a negative overall effect that reverses sign when a single outlier year (2016) is dropped. Across every design the data do not identify a causal effect of ShotSpotter on gun homicides in either direction, and they are inconsistent with a large deterrence effect of the kind a $3 million detection contract was expected to produce. The methodological lesson generalizes: a coefficient can survive fixed effects and robustness restrictions and still reflect specification choice and a design without a valid counterfactual.

1 Introduction

Acoustic gunshot detection promises to close a measurement gap. About one in five shootings is reported to police (Irvin-Erickson et al., 2016), so a sensor network that flags gunfire directly could speed response and, in principle, reduce deaths. Chicago acted on that premise. The city expanded ShotSpotter to 51 of its 77 community areas in 2017 and 2018 under a three-year, $3 million contract, then cancelled it in 2023 after an audit found that alerts rarely produced investigative evidence (Office of Inspector General, 2021).

The policy question is whether the program reduced gun violence—and for Chicago it is largely settled: district-level (Connealy et al., 2024) and national (Doucette et al., 2021) evaluations find no crime-reduction effect, so the contribution here is the identification assessment, not the null. The methodological question, which we treat as primary, is whether the available data can answer the policy question at all. We estimate the effect of ShotSpotter installation on gun homicides using a community-area-by-year panel for 2009 to 2023, and we hold each estimate against the assumptions its design requires. The headline difference-in-differences estimate is positive and significant. It does not survive correct modeling of the outcome, a test of parallel pre-trends, a placebo on pre-rollout years, or a check on the adequacy of the comparison group. The contribution is the demonstration itself: a worked case in which a coefficient that passes the usual robustness restrictions still fails the conditions for a causal interpretation.

2 Setting and data

ShotSpotter triangulates the sound of gunfire through fixed microphones, filters non-gunshot sounds, and forwards reviewed alerts to police (Choi, Librett, & Collins, 2014). Chicago concentrated coverage in its highest-violence neighborhoods. From 2009 to 2016, the 51 areas that later received ShotSpotter averaged 381 gun homicides per year; the 26 areas that never received it averaged 42, a ratio close to nine to one. Treatment was assigned where violence was already highest. That assignment rule is the source of every identification problem below.

The analysis combines four City of Chicago open-data products: the Violence Reduction victims file (61,450 incidents, 1991 to 2025), filtered to gun homicides (about 7,600 in 2009 to 2023); the historical ShotSpotter alert file (222,108 alerts, January 2017 to the program's termination on September 22, 2024; 191,824 fall in the 2017–2023 analysis window); the 2008 to 2012 Census socioeconomic indicators; and the community-area boundaries. The alert file shows a staggered rollout: 21 community areas activate in 2017 and 30 in 2018, which leaves 26 never-treated areas. Pooling both cohorts at a single 2017 cutoff, the convention in the original pipeline, biases the estimate when treatment timing varies (Goodman-Bacon, 2021); the cohort-aware estimators below address that bias.

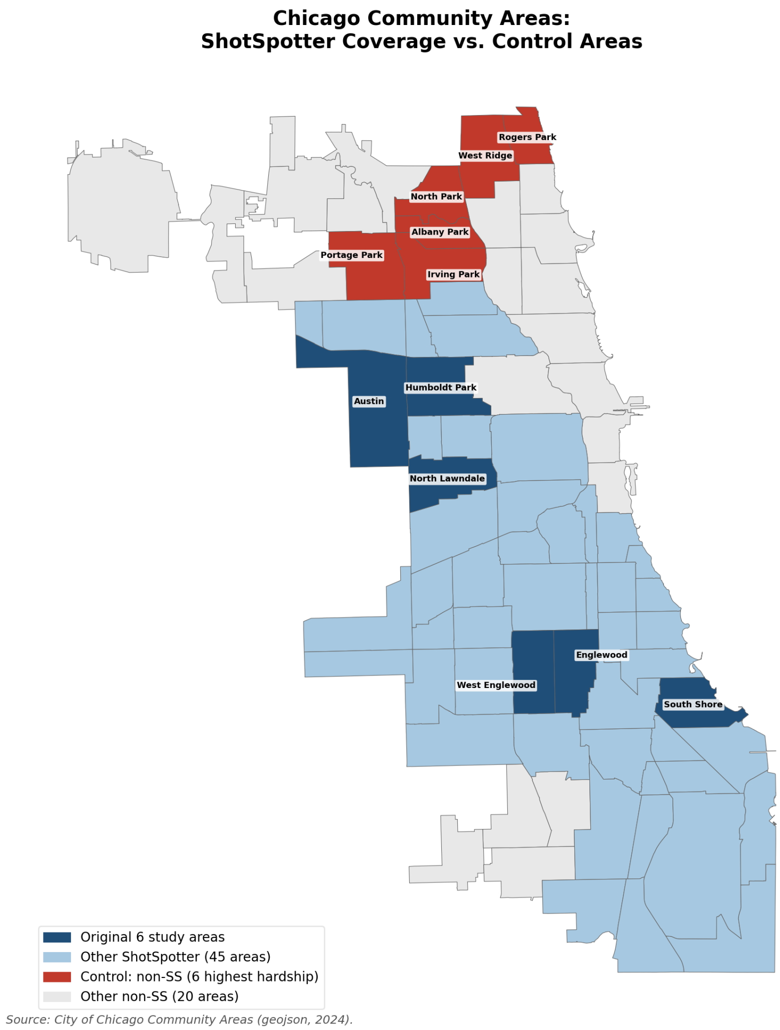

Figure 1: ShotSpotter coverage (blue) concentrates in the high-violence South and West sides. The 26 never-treated community areas form the comparison group, with the six highest-hardship areas shown in red. Treatment was assigned where violence was already highest.

3 Identification and threats to inference

Difference-in-differences identifies a causal effect only if treated and control units would have followed parallel paths absent treatment. Chicago's data strain that condition from the start. The treated areas began the period with roughly nine times the control areas' homicide rate, so the comparison rests on parallel trends rather than on comparable levels. We test the conditions the design needs and report each test against the threat it addresses (Table **??**).

Table 1: Conditions the difference-in-differences design requires, the diagnostic for each, and the result.

Condition required	Diagnostic	Finding	Status
Parallel pre-trends	Cohort event study; joint Wald F-test	$F = 19.9$, $p = 0.006$; pre-period effects large and negative	Rejected
Correct outcome model	Poisson and Negative Binomial against OLS	OLS $+2.63$ falls to IRR 1.07 ($p = 0.51$)	OLS misspecified
No spurious significance	Placebo treatment in pre-rollout years	1999 and 2003 significant at the 1% level	Design fails
Adequate comparison group	Synthetic control feasibility; permutation	Single-donor solution; pre-RMSE 10–34 against levels 15–70	Infeasible
Stability to staggered timing	Callaway–Sant'Anna against two-way FE	Overall effect reverses sign when 2016 is dropped	Not identified

4 Results

4.1 The headline estimate is a count-data artifact

The two-way fixed-effects regression of annual gun homicides on the treatment-by-post interaction, with community-area and year fixed effects and standard errors clustered by community area, returns $+2.627$ (SE 0.638, $p < 0.001$, 95% CI $[+1.38, +3.88]$, $N = 1{,}155$). The estimate is stable across restrictions: $+1.75$ excluding the COVID years 2020 and 2021, $+3.19$ excluding 2016, and $+2.31$ excluding both, each at $p < 0.001$. Adding the fixed effects reproduces the pooled coefficient almost exactly, so the result does not come from omitted area or year effects.

The estimate does come from the outcome model. Annual homicide counts are small non-negative integers whose variance rises with the mean. Ordinary least squares treats them as continuous and equal-variance and reports an additive effect. A Poisson or Negative Binomial regression with the same fixed effects lets the variance scale with the mean and reports a multiplicative effect, the natural scale for a proportional treatment. Estimated this way, the effect is statistically null in every specification (Table **??**). The Poisson incidence-rate ratio on the full panel is 1.07 (95% CI $[0.87, 1.33]$, $p = 0.51$); the Negative Binomial ratio is 1.07 (95% CI $[0.86, 1.33]$, $p = 0.55$), with the dispersion estimated at about 0.02 through the Cameron and Trivedi (2013) NB2 auxiliary regression rather than fixed at one. The interval admits anywhere from a 13% reduction to a 33% increase in homicides. On the additive scale the program looks consequential; on the multiplicative scale it is indistinguishable from zero (Figure **??**).

Table 2: Count-data difference-in-differences with two-way fixed effects. The incidence-rate ratio equals $\exp(\beta)$; a ratio of 1 indicates no effect.

Specification	β	IRR	95% CI on IRR	p
OLS (TWFE), full panel	+2.627	n/a	n/a	< 0.001
Poisson, full panel	+0.072	1.074	$[0.869, 1.328]$	0.508
Poisson, excl. COVID	+0.014	1.014	$[0.816, 1.259]$	0.902
Poisson, excl. 2016	+0.084	1.087	$[0.857, 1.381]$	0.491
Neg. Binomial, full panel	+0.067	1.069	$[0.860, 1.329]$	0.548
Neg. Binomial, excl. COVID	+0.014	1.014	$[0.810, 1.269]$	0.902
Neg. Binomial, excl. 2016	+0.074	1.077	$[0.844, 1.373]$	0.551

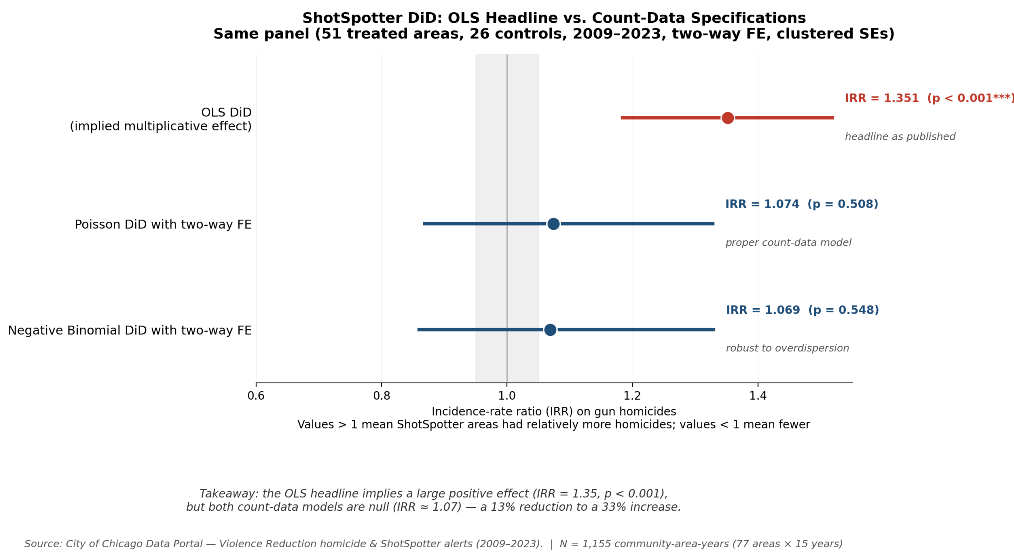

Figure 2: Treatment effect by estimator. The ordinary least squares coefficient appears on the multiplicative scale for comparison (implied incidence-rate ratio 1.35, $p < 0.001$). The Poisson and Negative Binomial points are estimated incidence-rate ratios, both statistically null near 1.07.

4.2 The design fails falsification

If the post-2017 estimate measured a treatment effect, the same regression with treatment assigned to years before ShotSpotter existed should return nothing. It does not. Placebo treatment in 1999 yields -1.34 ($p = 0.008$) and in 2003 yields -1.93 ($p = 0.007$), both significant at the 1% level and comparable in magnitude to the post-2017 estimate (Table **??**). The placebo coefficients are negative while the real estimate is positive. The opposite sign does not rescue the design; it shows that the specification detects pre-existing differences in neighborhood trajectories, which appear whatever year is labeled as treatment.

The stronger evidence is direction-free. A joint Wald test on the seven pre-treatment interaction coefficients of the event study rejects parallel trends ($F = 19.9$, $p = 0.006$). The cohort-aware event study makes the violation visible (Figure ??): every pre-treatment coefficient is large and negative, an artifact of 2016, a record homicide year concentrated in the soon-to-be-treated areas, sitting next to the base period. Once a design fails this test, the post-treatment contrast cannot be read as causal.

Table 3: Placebo tests. Treatment is reassigned to years before the ShotSpotter rollout.

Fake treatment year	β	SE	p	95% CI
1999	-1.344	0.507	0.008	$[-2.34, -0.35]$
2003	-1.929	0.714	0.007	$[-3.33, -0.53]$
2007	-0.645	0.497	0.194	$[-1.62, +0.33]$
2011	$+0.256$	0.347	0.461	$[-0.42, +0.94]$

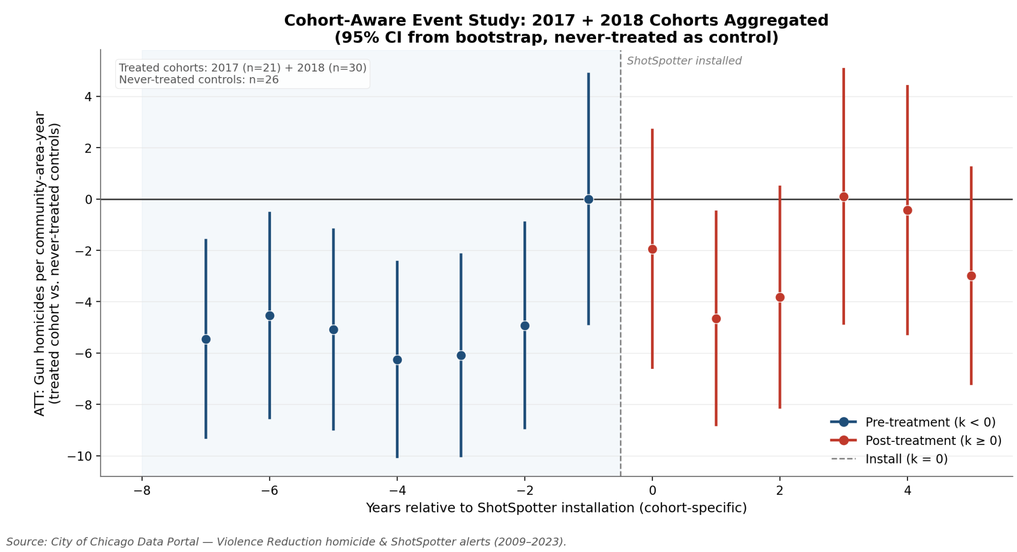

Figure 3: Cohort-aware event study with bootstrap confidence intervals. Large negative pre-treatment coefficients indicate a parallel-trends violation driven by the 2016 record homicide year adjacent to the base period.

4.3 No valid comparison group

A synthetic control (Abadie, Diamond, & Hainmueller, 2010) would build a counterfactual for each treated neighborhood from a weighted average of untreated areas. Here the construction fails. For all six study neighborhoods the optimizer places 100% of the weight on a single donor, Washington Heights, the most violent never-treated area, and the pre-treatment root mean squared error is 10 to 34 homicides per year against outcome levels of 15 to 70. The synthetic series sits well below the actual series even within the pre-treatment window it was fit to (Figure ??). Permutation inference returns a p-value below 0.001 for every treated unit, but that rejection follows from the level mismatch, not from a treatment effect. The never-treated areas occupy the low-violence end of the city, so no weighted combination reproduces a high-violence treated unit. The infeasibility is a property of the data: the city installed ShotSpotter in every high-violence neighborhood and left no comparable unit untreated.

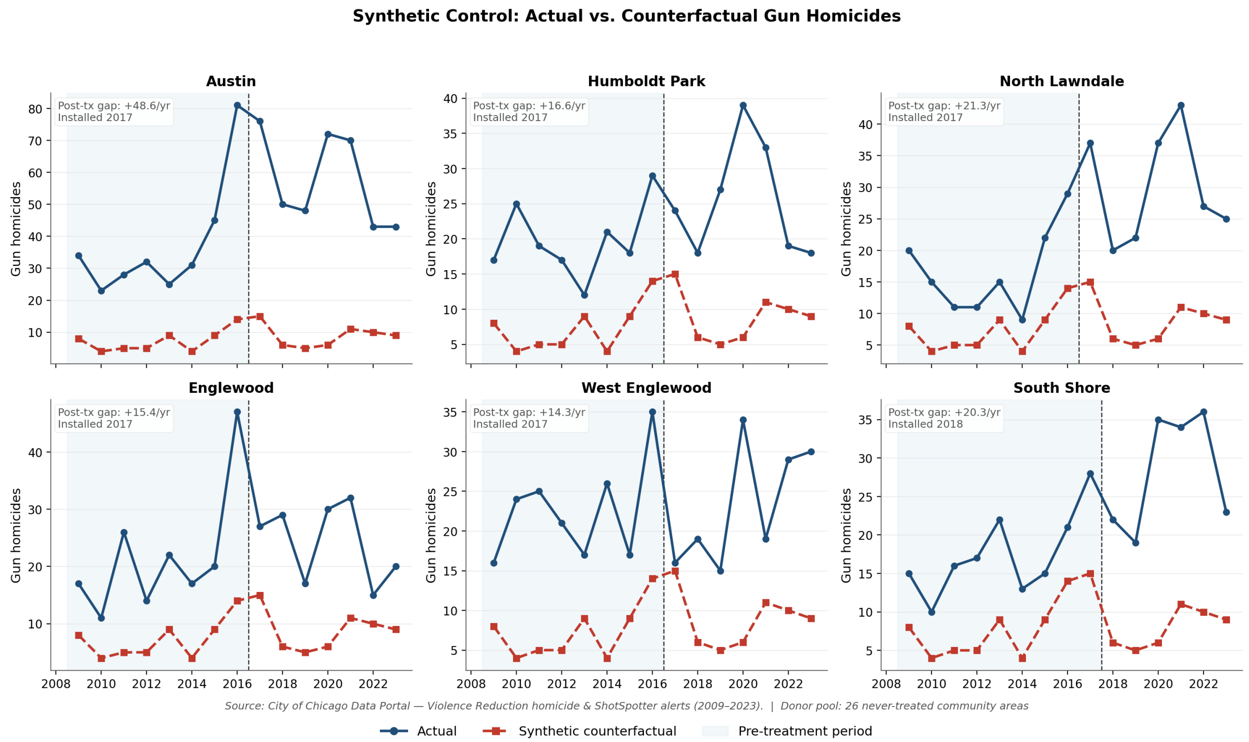

Figure 4: Synthetic control for the six study neighborhoods. The synthetic series (red) lies below the actual series (blue) even in the pre-treatment window it was fit to, because no combination of lower-violence never-treated areas matches a high-violence treated unit.

4.4 The modern estimator does not restore identification

The Callaway and Sant'Anna (2021) estimator handles staggered adoption without the negative weighting that biases two-way fixed effects under treatment-timing variation. On the full panel its overall effect is -2.45 per area-year (SE 0.89), negative and opposite in sign to the pooled regression, and stable to using not-yet-treated controls (-2.51, Table **??**). The sign is not a property of the estimator. It rests on the single 2016 pre-coefficient; the event study shows only that coefficient is significant while the rest sit near zero, and dropping 2016 returns $+1.90$ (SE 0.72), the sign of the pooled regression. Three estimators give three answers: $+2.63$ from ordinary least squares, a null ratio of 1.07 from the count models, and -2.45 reversing to $+1.90$ from Callaway and Sant'Anna. The spread is not estimator disagreement about a fixed parameter. It indicates a parameter the data do not pin down.

Table 4: Callaway and Sant'Anna (2021) overall effect, gun homicides per community-area-year.

Control group	Overall effect	SE
Never-treated	-2.453	0.893
Not-yet-treated	-2.508	0.898
Never-treated, excl. 2016	$+1.895$	0.724

5 What the data support

The results bound the answer without identifying it. The data do not show that ShotSpotter increased gun homicides, because the $+2.63$ estimate is a scale artifact. They do not show a reduction, because reading the negative event-study coefficients literally ignores that the base year is anomalous. The count-data confidence interval spans a 13% reduction to a 33% increase, the full policy-relevant range.

One conclusion survives. The data are inconsistent with a large *deterrence* effect. An intervention large enough to justify a $3 million detection contract would leave negative post-treatment coefficients with intervals excluding zero, stability across specifications, and a clearer signal in the higher-powered non-fatal-shooting series, which carries about 30,000 events. None of these appears. These data bound only the deterrence channel, however: a survival-channel mortality benefit of the size a boundary regression discontinuity attributes to faster emergency response (University of Chicago Crime Lab, 2024; not peer-reviewed) lies inside the count-model confidence interval and would be invisible in an area-by-year homicide panel. Whatever effect ShotSpotter had on homicides through deterrence was too small to separate from noise, from the 2016 anomaly, and from pre-existing trend differences across neighborhoods.

6 A better-identified design

A credible evaluation needs a counterfactual these data do not contain. Three designs would supply one. A geographic regression discontinuity at coverage boundaries would compare blocks just inside and just outside a ShotSpotter zone, holding neighborhood conditions roughly constant; it requires the sensor-coverage polygons, which the city has not released. Estimation at finer resolution, census tract by month rather than community area by year, would recover the within-neighborhood timing variation that the annual panel averages away. Direct measurement of the mechanism, police response time and time to trauma care around each activation date, would test the channel through which detection could reduce deaths instead of inferring it from counts. Each design targets the failure documented above, the absence of a comparable untreated unit.

7 Conclusion

ShotSpotter detected gunfire. Alert volume rose in covered areas because the sensors measure shots that 911 calls miss. The evaluation question is different, and these data answer it only in the negative. A significant difference-in-differences coefficient dissolves under a count model, a parallel-trends test, and a pre-rollout placebo, the comparison group cannot be constructed, and the modern staggered-adoption estimator changes sign on one year. With no clean counterfactual and a pre-period dominated by an outlier, the data cannot identify the effect of ShotSpotter on gun homicides in either direction. The 2023 decision to end the contract is consistent with that absence of evidence. The wider lesson carries to any program evaluated this way: a coefficient that survives fixed effects and robustness restrictions can still fail the conditions a causal reading requires.

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